Summary

Here, we explore two prominent numerical methods for solving ordinary differential equations: the Bulirsch-Stoer method and the fourth-order Runge-Kutta method. We enhance their implementation by incorporating adaptive step size control in order to optimize computational efficiency while maintaining a desired level of accuracy. We first apply the Bulirsch-Stoer method to the Belousov-Zhabotinsky (BZ) reaction, an example of a chemical oscillator, and then use the fourth-order Runge-Kutta method to integrate the orbit of a supermassive black hole binary in the presence of dynamical friction. We find that adaptive integration accurately captures the oscillatory behavior of the BZ reaction and that dynamical friction allows the black hole binary to reach the Schwarzschild radius, with the inspiral time depending on the initial velocity and the strength of dynamical friction.

Methods

Bulirsch-Stoer Method

When given a differential equation of the form $\frac{dx}{dt} = f(x,t)$ with initial conditions, the Bulirsch-Stoer method provides a numerical solution over a specified interval (t to $t+H$). This requires first employing the modified midpoint method to estimate the solution at $t+H$ using n substeps of size $h = H/n$. Letting x_m and y_m denote the solution at integer and half-integer multiples of h, respectively, we can define:

$$y_{m+1} = y_m + hf(x_m, t+mh), \quad x_{m+1} = x_m + hf(y_{m+1}, t+(m+\frac{1}{2})h)$$

with the concluding two points in the solution being $y_n = x(t+H-\frac{1}{2}h)$ and $x_n = x(t+H)$. Using y_n, we can obtain another estimate of $x(t+H)$ by utilizing Euler's method with a step size of $-\frac{1}{2}h$ to get $x(t+H-\frac{1}{2}h) = x(t+H) - \frac{1}{2}hf(x(t+H), t+H)$. Rewriting this in terms of y_n and x_n, we get $y_n = x(t+H) - \frac{1}{2}hf(x_n, t+H)$. Rearranging this equation, our secondary estimate of $x(t+H)$ becomes $x(t+H) = y_n + \frac{1}{2}hf(x_n, t+H)$. Combining this estimate with the original $x(t+H) = x_n$ by averaging, we arrive at the formula for the modified midpoint method:

$$x(t+H) = \frac{1}{2}(x_n + y_n + \frac{1}{2}hf(x_n, t+H)).$$

The Bulirsch-Stoer method solves the differential equation by applying the modified midpoint method from t to $t+H$ with successive values of increasing n. Let the current number of steps be denoted as n and the modified midpoint estimate of $x(t+H)$ be $R_{n,1}$. One can then increase the accuracy of the estimate by leveraging the fact that the total error on the modified midpoint method is an even function of the step size. Recognizing this allows us to recursively cancel leading error terms by combining estimates at different values of n using Richardson extrapolation ($R_{n,m+1}$). Thus,

$$x(t+H) = R_{n,m+1} + O(h_n^{2m+2}), \quad R_{n,m+1} = R_{n,m} + \frac{R_{n,m} - R_{n-1,m}}{[n/(n-1)]^{2m} - 1}.$$

This reduces the number of modified midpoint steps required to achieve a desired accuracy as $R_{n,1}$ is computed as the modified midpoint approximation, which is then supplemented by

further extrapolations which each depends on the previous two estimates and goes on until the desired accuracy is achieved. The Bulirsch-Stoer method has two main limitations: (1) it calculates an accurate valye for the final value of $x(t+H)$, but all intermediate values are only the midpoint estimates which aren't as accurate, and (2) since we are calculating terms in a series expansion, the method is vulnerable to time bottlenecks if the series is slow to converge (therefore, it is usually practical to keep the number of modified midpoint steps to a maximum of 8 to 10). These limitations can be addressed by dividing our time interval into smaller subintervals of size $(b-a)/N = H$ for initial and final time of a and b, respectively, and N subintervals. The Bulirsch-Stoer method can then be applied to each subinterval, which allows for more accurate intermediate values and reduces the time bottleneck by keeping the number of modified midpoint steps small.

Ergo, the Bulirsch-Stoer method can be summarized as the following algorithm from Newman's *Computational Physics*:

1. In just one step, solve the differential equation from t to $t+H$ using the modified midpoint method to get $R_{1,1}$.
2. Successively increase n by one to get new estimates of the modified midpoint $R_{n,1}$ using n steps.
3. Use the Richardson extrapolation formula to get $R_{n,m}$ to get more estimates.
4. After calculating a whole set of $R_{n,m}$ for constant n, compare at the end the error given by $c_m h_n^{2m} = \frac{R_{n,m}-R_{n-1,m}}{[n/(n-1)]^{2m}-1}$ to see if it is close to the desired accuracy target. If the error is still too big, go back to step 2, otherwise go onto the next time interval H.

Adaptive Scheme for Automatic Step Size Control

As discussed above, the Bulirsch-Stoer method works best when n is minimized since we want to limit how many modified midpoint steps are calculated, ideally capping n at 8 to 10 at any recursive step of the calculation. To do this, one must make a suitable selection of H so that it is small enough for convergence towards the target accuracy to be speedy. Instead of having H be wholly user defined, we can implement an automatic step size control to be executed by the computer, which is particularly useful for hefty calculations. We can therefore implement an adaptive scheme to automatically adjust the step size H based on the error estimate of the previous step.

Implementation is as follows: Once again solving the differential equation from $t = a$ to $t = b$, we can divide this time interval by some small number N (perhaps something from 1 to 4) so that the size of each subinterval is $(b-a)/N$. The Bulirsch-Stoer method is then to be applied on each of the interval as outlined above, each interval being subdivided into n midpoint steps and extrapolating as we increase n. Each value of n gets an error estimate, and if accuracy reaches the target, then that interval is complete and we move onto the next. However, if n gets to the maximum allowed value and we are not accurate enough yet, we no longer continue with the calculation and instead subdivide the current time interval of size H to $H/2$ and repeat the process. This is done until the target accuracy is reached. Any interval that doesn't reach the target accuracy before maximum n is reached will be subdivided into two parts and the process is applied to each of the two new subintervals separately. Thus, we are basically refining the resolution of the step size in an equitable manner given what each

subinterval requires to reach the desired accuracy. This is a very efficient way to implement the Bulrisch-Stoer method as we are not wasting time on subintervals that are already accurate enough.

Fourth-Order Runge-Kutta Integrator with Adaptive Time Steps

The Runge-Kutta method is a family of numerical methods for solving ordinary differential equations. In the previous subsection on the Bulirsch-Stoer method, we fleetingly discussed Euler's method, which is the first-order Runge-Kutta method, and the modified midpoint method, a related second-order method that extends the Runge-Kutta midpoint approach by using leapfrog steps, in which the derivative evaluated at an intermediate point advances the solution directly from the preceding point to the following one across an interval of width $(2h)$. Runge-Kutta methods improve accuracy by evaluating the derivative $f(x,t)$ at multiple points within each time step, rather than relying solely on its value at the beginning of the interval. One of the more common Runge-Kutta methods is the fourth-order version (RK4), which estimates the solution of $\frac{dx}{dt} = f(x,t)$ over step size h using four intermeidate quantities:

$$\begin{aligned} k_1 &= hf(x,t), \\ k_2 &= hf(x + \frac{1}{2}k_1, t + \frac{1}{2}h), \\ k_3 &= hf(x + \frac{1}{2}k_2, t + \frac{1}{2}h), \\ k_4 &= hf(x + k_3, t + h), \\ x(t+h) &= x(t) + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

By estimating the change in x through its derivatives at the beginning, midpoint, and end of the interval, the RK4 estimates the solution using the weighted avergae in the last equation above, achieving an accuracy of order h^4 and with an error of order h^5.

To further improve the computational efficiency of the RK4 method, we can implement adaptive time stepping, wherein the step size h is automatically adjusted depending on how rapidly the solution changes. Rather than using a fixed step size throughout the entire integration, we can estimate the local truncation error by comparing two different RK4 approximations over the same time interval. Following the procedure outlined in the lecture notes, let x_1 be the estimate obtained by taking one large RK4 step of size $2h$ from t to $t + 2h$, and x_2 be the estimate obtained by taking two smaller RK4 steps of size h over the same interval. Since the local truncation error of RK4 is proportional to h^5, the error in the large step is approximately $c(2h)^5 = 32ch^5$, while the accumulated error in the two smaller steps is approximately $2ch^5$, where c is a coefficient depending on the solution. Thus, the difference between the two estimates is approximately $30ch^5$, allowing us to estimate the error associated with a single smaller step as

$$\epsilon = ch^5 \approx \frac{1}{30}|x_1 - x_2|.$$

Now, letting δ denote our desired accuracy per unit time, the target accuracy for a single step of size h is δh. We therefore want to choose an optimal step size h_0 such that the

estimated error does not exceed this target. Recognizing that the local truncation error scales as the fifth power of the step size, we can write

$$\delta h_0 = ch_0^5 = \frac{|x_1 - x_2|}{30}\left(\frac{h_0}{h}\right)^5.$$

Rearranging this equation, we obtain the optimal step size

$$h_0 = h\left(\frac{30h\delta}{|x_1 - x_2|}\right)^{1/4} = h\rho^{1/4}, \qquad \rho = \frac{30h\delta}{|x_1 - x_2|}.$$

Here, ρ represents the ratio between the desired accuracy and the estimated error. If $\rho \geq 1$, the two smaller RK4 steps satisfy our accuracy requirement, and we accept x_2 as our solution at $t+2h$. Otherwise, the step is rejected and we repeat the calculation with a smaller h. To avoid choosing a step size that is too large and having to reject it repeatedly, we also introduce a safety factor of 0.9 when updating h. Thus, the step size is adjusted according to

$$h_{\text{new}} = 0.9h\rho^{1/4}.$$

In practice, we also limit how much the step size can increase or decrease in a single iteration. This process is repeated throughout the integration, allowing us to take smaller time steps where the solution changes rapidly and larger time steps where the solution varies more slowly. In this way, the adaptive RK4 method maintains our desired accuracy without unnecessarily increasing the number of calculations.

Results

Oscillating Chemical Reactions: The Belousov-Zhabotinsky Reaction

The Belousov-Zhabotinsky (BZ) reaction is a classic example of a chemical oscillator, where a heated solution of chemical experience a set of reactions that lead to the chemical concentrations to oscillate between extremes. As long as you heat the mixture of chemicals, the oscillation will continue, which can be visualized using color dye. The Physicist Ilya Prigogine modeled this reaction mathematically using a pair of ordinary differential equations that describe the time evolution of the concentrations of chemicals x and y:

$$\frac{dx}{dt} = 1 - (b+1)x + ax^2y, \quad \frac{dy}{dt} = bx - ax^2y.$$

This system of equations was coined the "Brusselator" model, paying homage to the city of Brussels where Prigogine worked. Here, we apply the Bulirsch-Stoer method with adaptive step size control to solve the Brusselator for the case of $a = 1$ and $b = 3$ with the initial conditions $x_0 = y_0 = 0$ to an accuracy of at least $\delta = 10^{-10}$ per unit time in x and y over the time interval $t \in [0, 20]$. We use a maximum of $n = 8$ modified midpoint steps before the adaptive step size control kicks in. The results are shown in fig. 1.

We can clearly see in the plot that the solution points are clustered where the variables are changing rapidly (namely, the peaks and troughs of the oscillations).

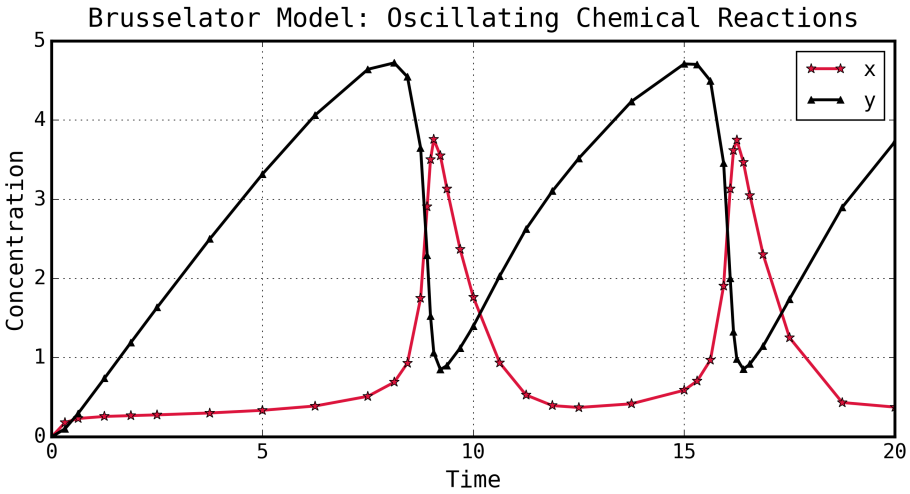


Figure 1: The concentrations of chemicals x and y in the Brusselator model over time, solved using the Bulirsch-Stoer method with adaptive step size control. The oscillatory behavior of the chemical concentrations is clearly manifest, as exhibited by the periodic fluctuations in both x and y.

Supermassive Black Hole Binary

Next, we use the fourth-order Runge-Kutta method with adaptive time steps to integrate the orbit of a supermassive black hole binary in the presence of dynamical friction. A black hole has the following equation of motion:

$$\frac{d^2\vec{r}_{\rm BH}}{dt^2} = -\frac{GM_{\rm BH}}{4r_{\rm BH}^3}\vec{r}_{\rm BH} + \dot{\vec{v}}_{\rm DF},$$

where $\dot{\vec{v}}_{\rm DF}$ is the acceleration due to dynamical friction and can be approximated as $\dot{\vec{v}}_{\rm DF} = -\frac{A}{v_{\rm BH}^3+B}\vec{v}_{\rm BH}$, with A representing the product of the black hole mass and stellar density, and B representing the velocity dispersion of the stellar field cubed. We seek to determine whether the black hole binary can get close enough at pericenter to lose energy through gravitational radiation, i.e. $r_{\rm pericenter} \sim r_s$, where r_s is the Schwarzschild radius. We set $M = G = 1$ and take the unit of distance to be 100 parsecs, such that $r_s = 10^{-7}$ in our numerical units. Since the orbit is symmetric, we only need to solve for the position of one black hole. For calculations with dynamical friction, our initial conditions are $x_0 = 1$, $y_0 = 0$, and an initial tangential velocity equal to 0.8 times the circular velocity. Since the black holes have equal masses and are separated by a distance $2r$, we can equate the gravitational acceleration to the centripetal acceleration:

$$\frac{GM}{(2r)^2} = \frac{v_{\rm circ}^2}{r} \quad \Longrightarrow \quad v_{\rm circ} = \sqrt{\frac{1}{4r}}, \qquad v_0 = 0.8v_{\rm circ}.$$

(a)

Let's first solve for the orbit without dynamical friction, with the initial velocity such that $r_{\text{pericenter}} = 10^{-7}$. Without dynamical friction, our equation of motion is $\ddot{\vec{r}} = -\frac{\vec{r}}{4r^3}$. For a Keplerian orbit, the vis-viva equation is $v^2 = GM(\frac{2}{r} - \frac{1}{a})$. In our case, the effective gravitational parameter is $GM/4 = 1/4$, and the semimajor axis is $a = (r_{\text{peri}} + r_{\text{apo}})/2$. Choosing an initial position at the apocenter $r_{\text{apo}} = 1$, we can therefore calculate the initial tangential velocity as

$$v_0^2 = \frac{1}{4}\left(\frac{2}{r_{\text{apo}}} - \frac{1}{a}\right) = \frac{r_{\text{peri}}}{2r_{\text{apo}}(r_{\text{apo}} + r_{\text{peri}})},$$
$$v_0 = \sqrt{\frac{10^{-7}}{2(1)(1+10^{-7})}} \approx 2.236 \times 10^{-4}.$$

Thus, we take the initial conditions to be $(x_0, y_0, v_{x,0}, v_{y,0}) = (1, 0, 0, 2.236 \times 10^{-4})$. The orbital period is $T = 2\pi\sqrt{4a^3} \approx 4.443$ in our numerical units, so we integrate the orbit for 12 periods.

To determine an appropriate error tolerance per unit time, δ, we compare one fourth-order Runge-Kutta step of size $2h$ with two steps of size h, as discussed in the methods above. The difference between the two position estimates gives our error estimate, and we adjust the time step accordingly. Since energy and angular momentum should be conserved in the absence of dynamical friction, we calculate

$$E = \frac{1}{2}(v_x^2 + v_y^2) - \frac{1}{4r}, \qquad L = xv_y - yv_x,$$

and compare their values throughout the integration. We also calculate the pericenter from the orbital energy and angular momentum to check whether it remains approximately constant.

We test $\delta = 10^{-8}$, 10^{-9}, and 10^{-10}, finding that decreasing the tolerance improves the conservation of energy, angular momentum, and pericenter. As shown in Fig. 2, for $\delta = 10^{-10}$, the maximum relative energy error is approximately 3×10^{-6}, while the angular momentum error and relative pericenter drift remain below 2×10^{-11} and 4×10^{-11}, respectively. The energy error increases most noticeably near each pericenter passage, where the black hole moves fastest and the gravitational acceleration changes most rapidly. Nevertheless, there is no appreciable loss of accuracy over 12 orbits, so we choose $\delta = 10^{-10}$ for the remaining calculations.

(b)

Next, we include dynamical friction by setting $A = B = 1$. Using the specified initial conditions, $x_0 = 1$, $y_0 = 0$, and $v_0 = 0.8v_{\text{circ}}$, we have $v_{\text{circ}} = \sqrt{1/4} = 0.5$, so that $(v_{x,0}, v_{y,0}) = (0, 0.4)$. We integrate the orbit using the same adaptive Runge-Kutta method and stop when the black hole reaches the Schwarzschild radius, $r_s = 10^{-7}$.

As shown in Fig. 3, the black hole follows a progressively smaller orbit as dynamical friction removes energy and angular momentum. In particular, the successive pericenter passages become smaller until the black hole reaches the Schwarzschild radius. We find that this occurs at $t \approx 10.755646$ numerical time units, after 150408 accepted integration steps.

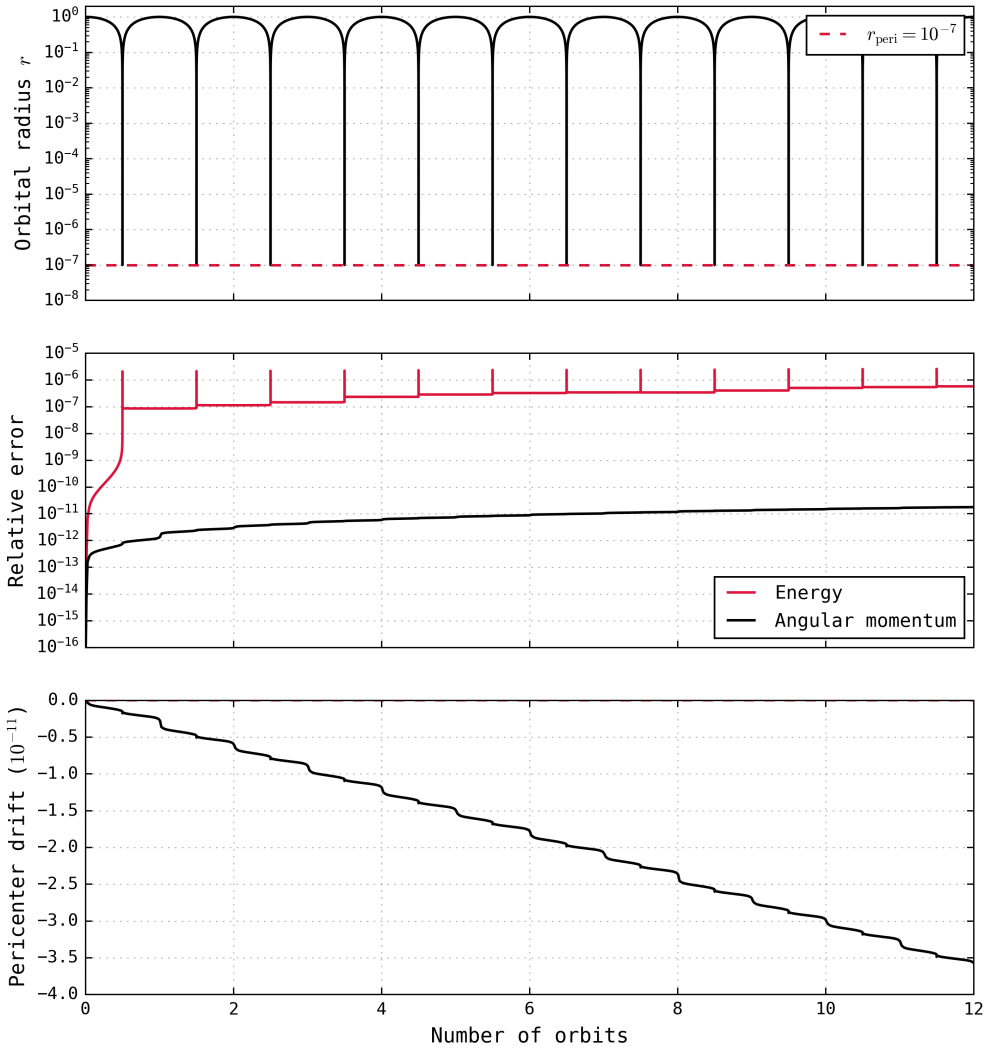


Figure 2: Orbital radius, relative errors in energy and angular momentum, and relative pericenter drift over 12 orbital periods without dynamical friction, using $\delta = 10^{-10}$.

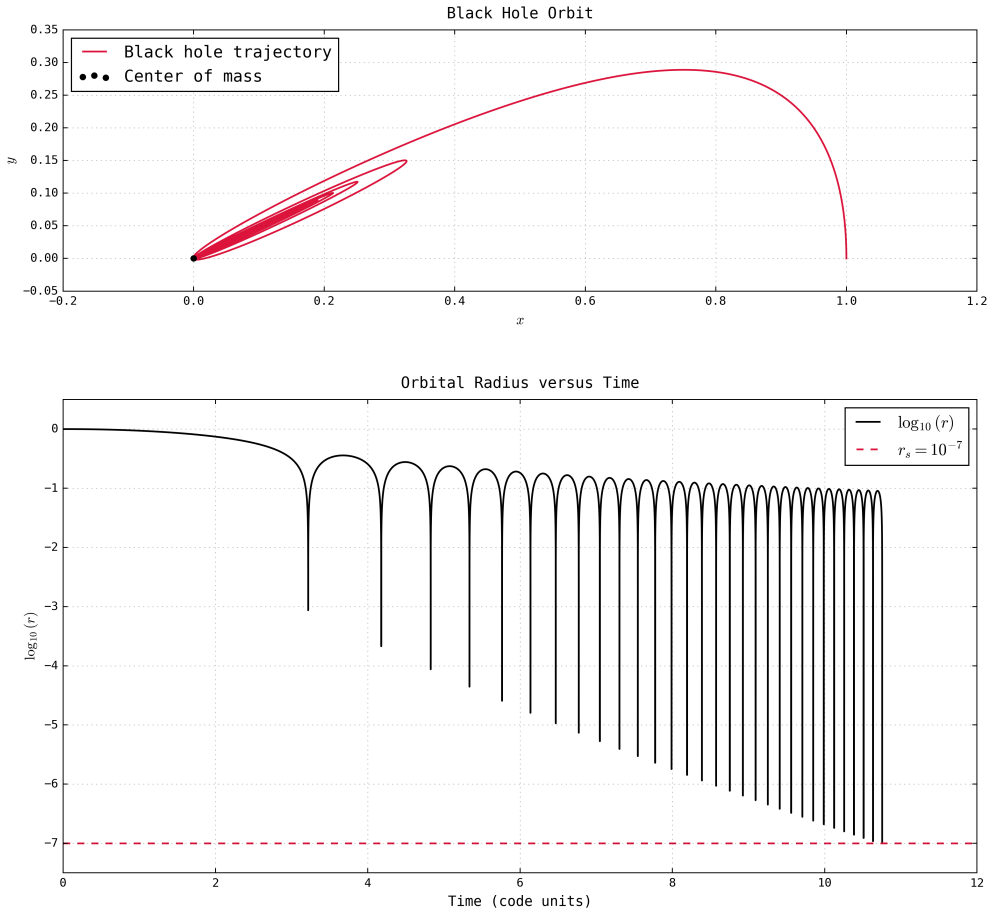


Figure 3: Black hole trajectory in the presence of dynamical friction with $A = B = 1$ (top), and $\log_{10}(r)$ as a function of time (bottom). The dashed red line indicates the Schwarzschild radius, $r_s = 10^{-7}$.

Thus, for $A = B = 1$, the binary can reach the separation scale at which energy loss through gravitational radiation becomes important.

(c)

We now investigate how the time required to reach the Schwarzschild radius depends on B/A, varying both A and B over the interval $[0.5, 10]$. To convert our numerical time to Myr, we use the fact that the distance unit is $R_0 = 100\,\mathrm{pc}$ and that $r_s = 10^{-7}$ in these units. Assuming that r_s corresponds to the Schwarzschild radius of an individual black hole, we have

$$r_s^{\mathrm{phys}} = 10^{-7}(100\,\mathrm{pc}) = 10^{-5}\,\mathrm{pc}, \qquad M_{\mathrm{BH}} = \frac{c^2 r_s^{\mathrm{phys}}}{2G} \approx 1.05 \times 10^8\, M_\odot.$$

The corresponding unit of time is therefore

$$t_0 = \sqrt{\frac{R_0^3}{GM_{\mathrm{BH}}}} \approx 1.4586\,\mathrm{Myr}.$$

Thus, the inspiral time found in part (b) corresponds to approximately $10.755646(1.4586) = 15.6884\,\mathrm{Myr}$.

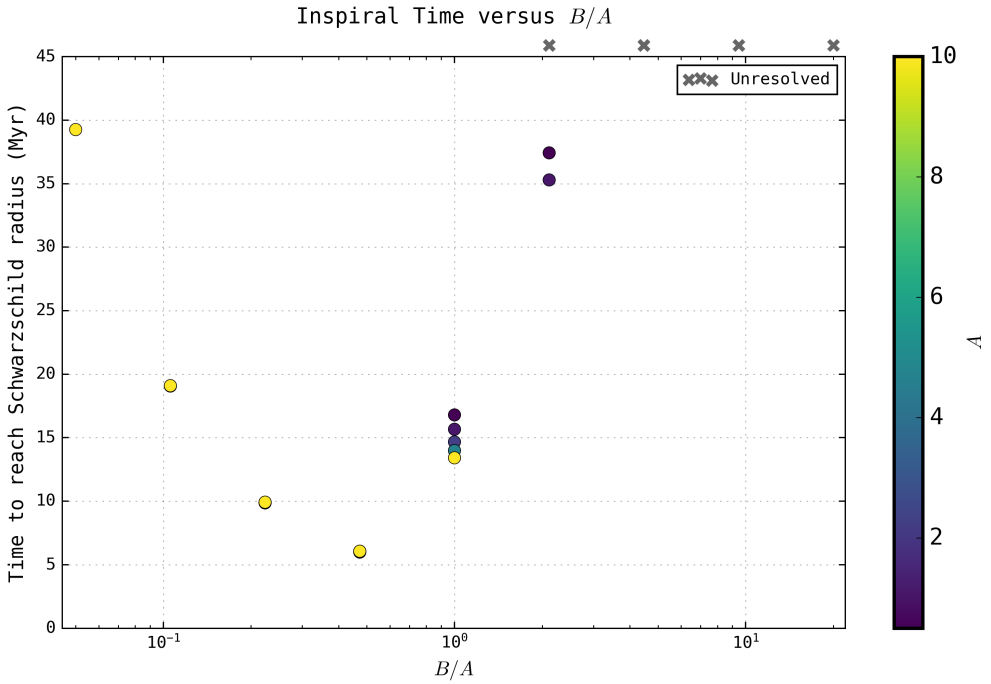


Figure 4: Time required for the black hole to reach the Schwarzschild radius as a function of B/A, for $A, B \in [0.5, 10]$. The colors indicate the individual values of A, while the crosses at the top mark calculations that did not resolve a crossing within the numerical integration limits.

As shown in Fig. 4, the inspiral time is not a monotonic function of B/A. For the parameter combinations we tested, the shortest times occur around $B/A \approx 0.47$, where the black hole reaches r_s in approximately 6 Myr. The inspiral time becomes longer both for smaller and

larger ratios. For example, at $B/A = 0.05$, we obtain approximately 39.26 Myr, while at $B/A \approx 2.11$, the resolved calculations give approximately 35–37 Myr. Some combinations with larger ratios could not be resolved within our computational limits, so we cannot assign them a definite inspiral time.

We can understand this behavior qualitatively from the dynamical friction acceleration,

$$\dot{\vec{v}}_{\rm DF} = -\frac{A}{v^3 + B}\vec{v}.$$

When $v^3 \ll B$, the acceleration is approximately $-(A/B)\vec{v}$. Thus, for large B/A, dynamical friction is weaker, so the black hole loses orbital energy more slowly and takes longer to reach the Schwarzschild radius. On the other hand, when B/A is very small, friction becomes sufficiently strong that it can substantially suppress the black hole's radial motion, increasing the time required for it to move inward. Consequently, intermediate values of B/A can produce the shortest inspiral times.

(d)

Finally, we investigate whether the inspiral time depends on the initial velocity and whether it is determined solely by B/A or also by the individual values of A and B.

First, keeping $A = B = 1$, we vary the initial tangential velocity between $0.6v_{\rm circ}$ and $0.95v_{\rm circ}$. As shown in the upper panel of Fig. 5, the inspiral time increases from approximately 14.95 Myr to 16.06 Myr. Thus, our results do depend on the initial velocity. A larger initial tangential velocity gives the black hole greater initial angular momentum and changes its orbital energy, affecting the time required to reach the Schwarzschild radius.

Next, we hold $B/A = 1$ constant while varying $A = B$ from 0.5 to 10. As shown in the lower panel of Fig. 5, the inspiral time decreases from approximately 16.79 Myr to 13.41 Myr, even though the ratio B/A remains unchanged.

We can see why the individual values matter by rewriting the dynamical friction acceleration as

$$\dot{\vec{v}}_{\rm DF} = -\frac{1}{v^3/A + B/A}\vec{v}.$$

Even if B/A is fixed, the term v^3/A changes when A changes. Therefore, the magnitude of dynamical friction is not generally determined by B/A alone. We conclude that the time required for the black hole to reach the Schwarzschild radius depends on both its initial velocity and the individual values of A and B, rather than only on their ratio.

Code

The link to the github repository for this homework assignment can be accessed by clicking here.

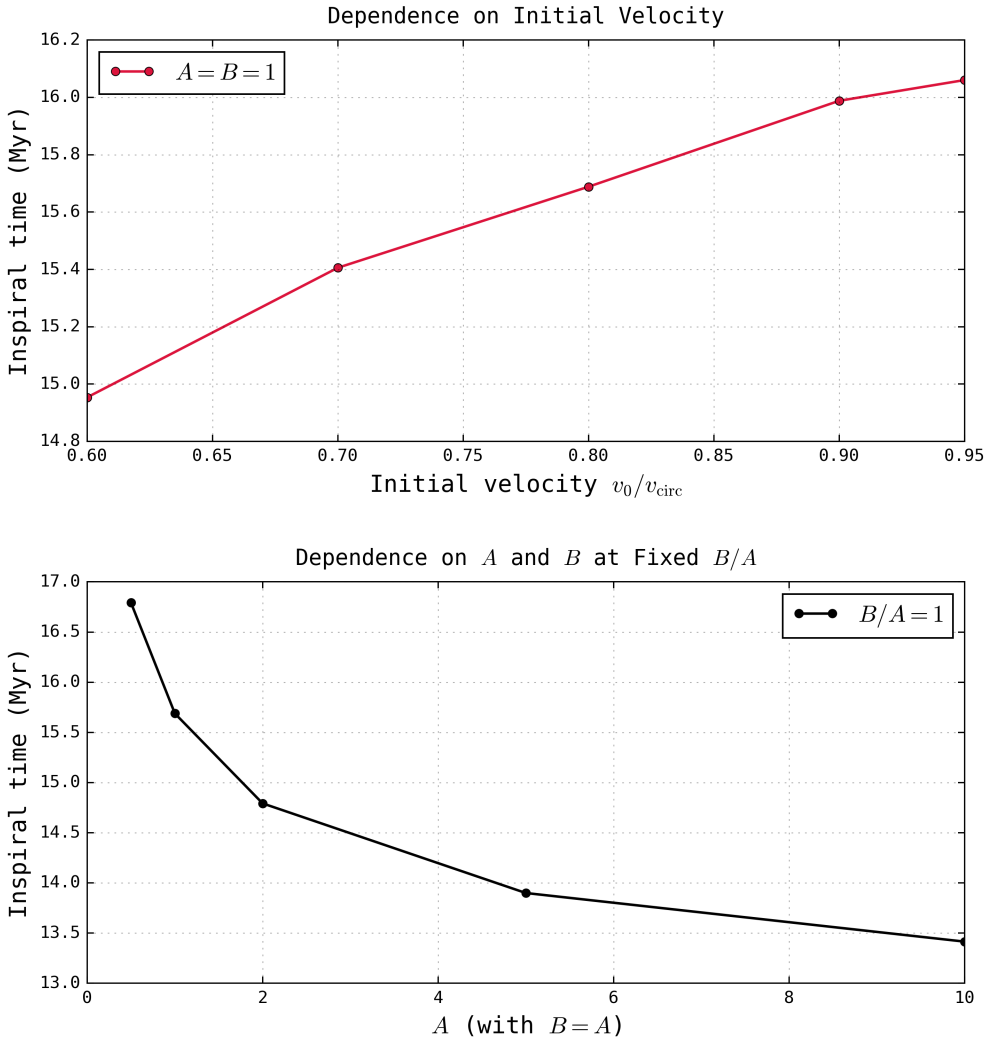


Figure 5: Dependence of inspiral time on the initial velocity for $A = B = 1$ (top), and on the individual values of A and B when their ratio is fixed at $B/A = 1$ (bottom).